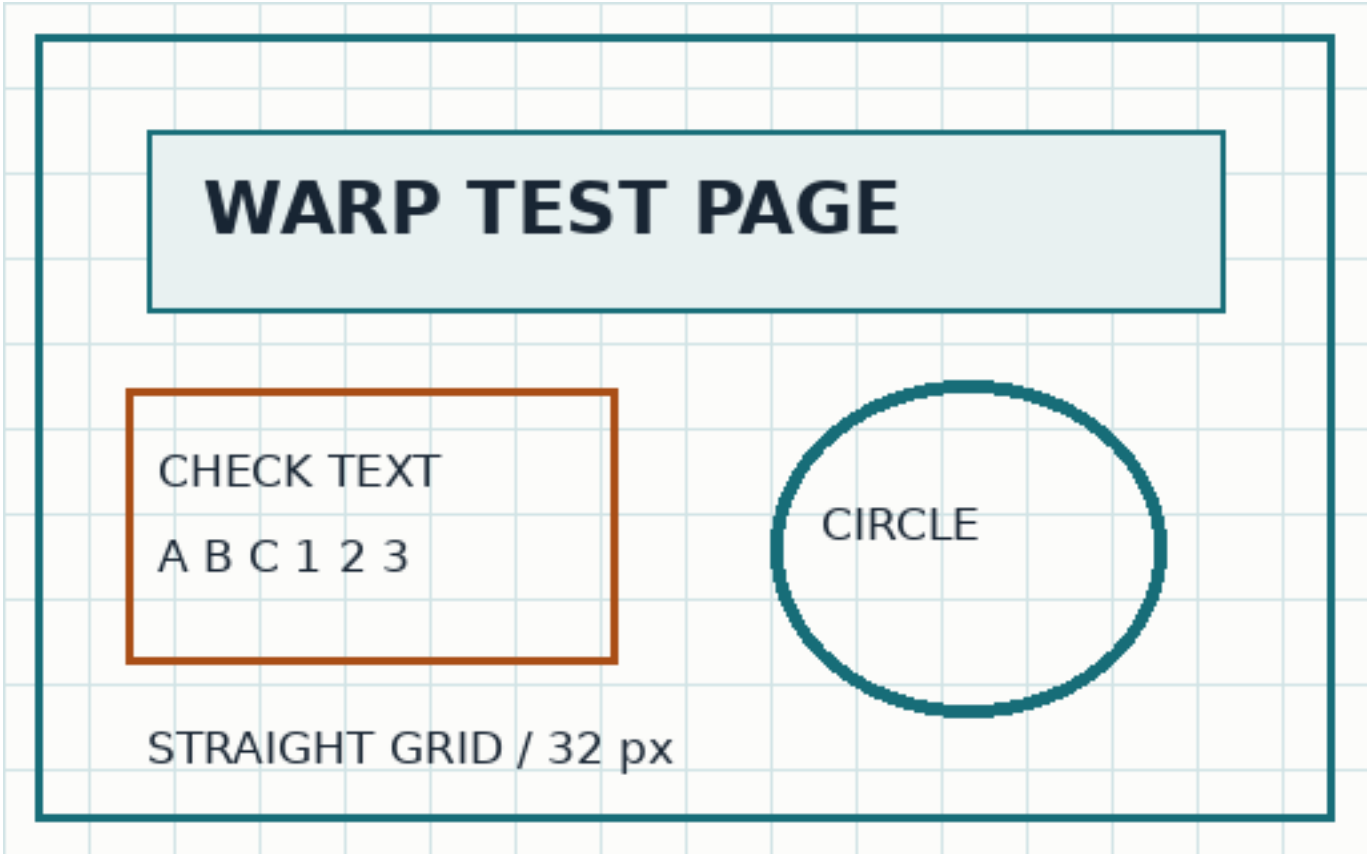
# IMAGE WARPS

How the effects change with intensity and thickness




SOURCE IMAGE / 512 × 320 pixels

The grid and letters show direction, local shape, and text legibility.

All comparison images come from the installed lorem_warped_examples code.

## READING ORDER

1 Read the field and scale guide.

2 Compare seven effect sweeps.

3 Use the Python or CLI examples.

# Read the images

Keep these units separate when you compare the effects.

### 01 INPUT

A Python image has CHW order. The examples use one 512 × 320 RGB image.

### 02 FIELD

The field stores (dy, dx) at each output pixel. Positive values move content down or right.

### 03 INTENSITY

Geometric modes multiply one fixed field by an intensity in pixels.

### 04 THICKNESS

Refraction uses slab thickness in mm. It does not use artistic intensity.

### BACKWARD SAMPLE

output(y, x) = input(y - intensity × dy, x - intensity × dx)

The sampler uses bilinear interpolation and repeats border pixels.

## USE CASES

Document augmentation: waves, droplets, and lens cells.

Grid inspection: see the direction of geometric displacement.

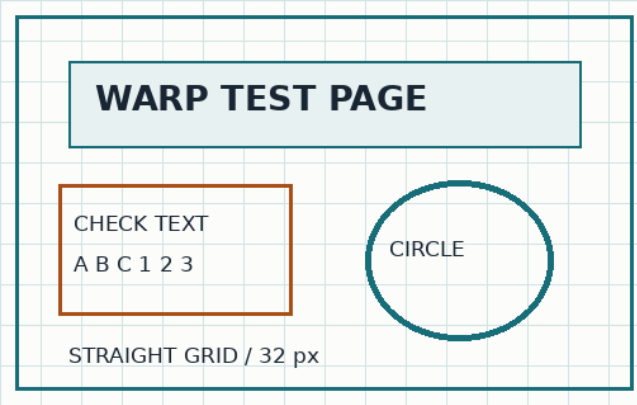
Optical view: compare two mode translations through a slab.

EFFECT SCALE / HORIZONTAL_WAVE

# Horizontal wave

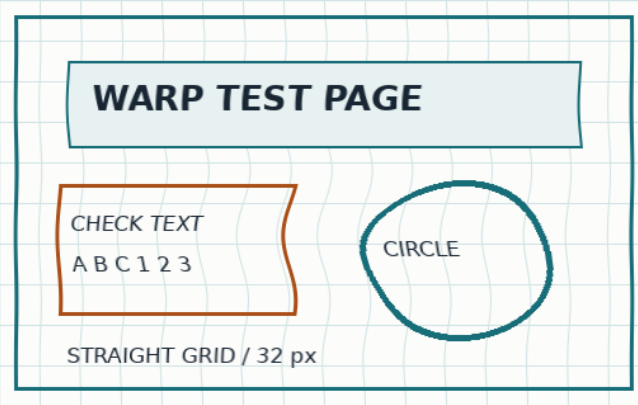
Intensity changes one fixed field.

### A = 0 px



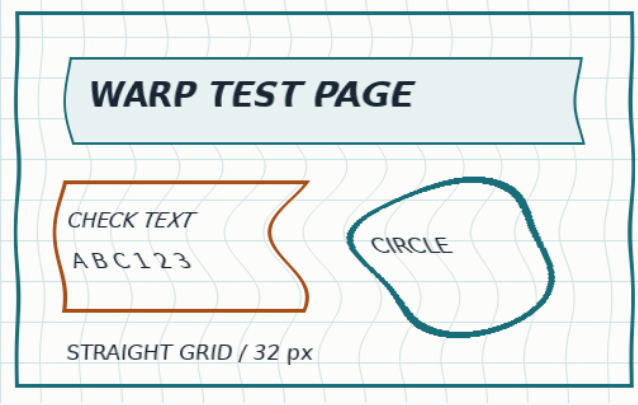

max field shift: 0.0 px | mean change: 0.000

### A = 6 px



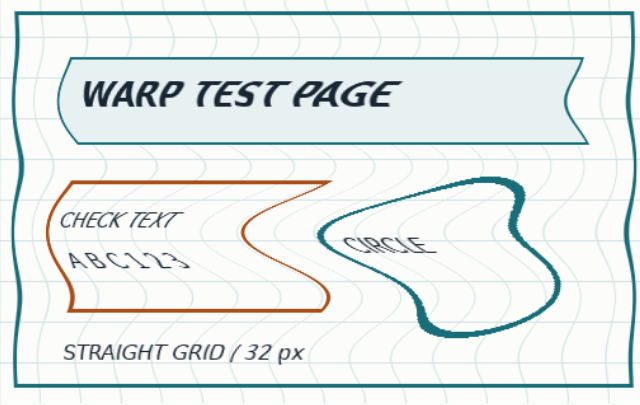

max field shift: 5.8 px | mean change: 0.030

### A = 18 px




max field shift: 17.4 px | mean change: 0.044

### A = 42 px




max field shift: 40.6 px | mean change: 0.052

## HOW IT WORKS

The field moves visible content left and right. It keeps page edges fixed.

Use this mode to test lateral line movement in a text image.

**LIMIT:** At high intensity, letters can overlap.

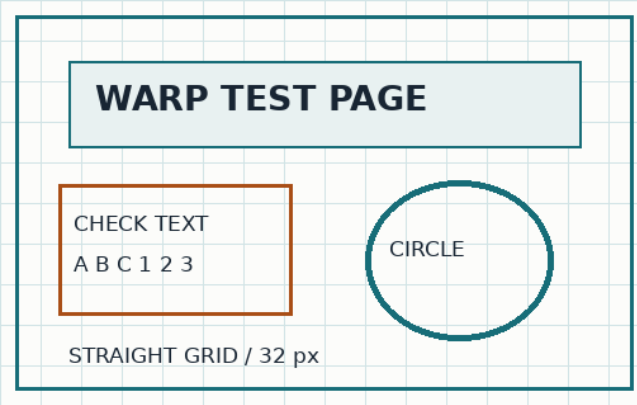
```
WarpTransform(mode="horizontal_wave", intensity=18)(image)
```

EFFECT SCALE / VERTICAL_WAVE

# Vertical wave

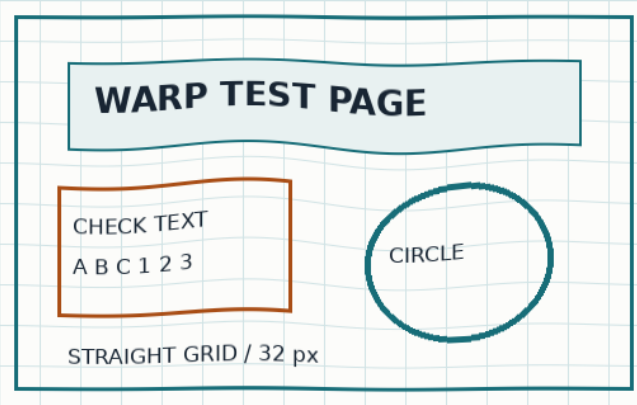
Intensity changes one fixed field.

### A = 0 px



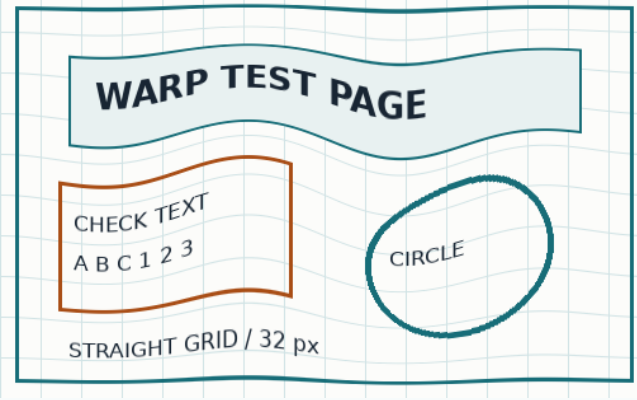

max field shift: 0.0 px | mean change: 0.000

### A = 6 px



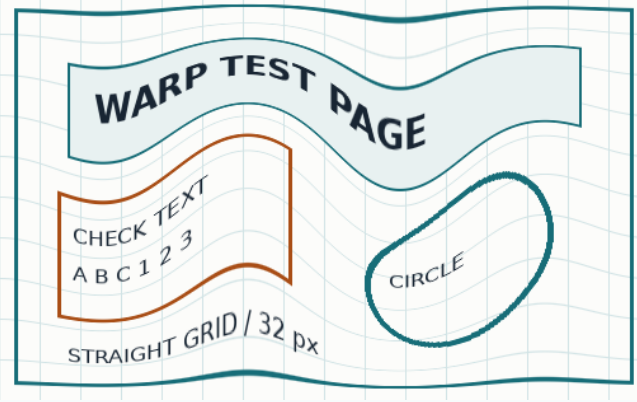

max field shift: 5.6 px | mean change: 0.040

### A = 18 px




max field shift: 16.7 px | mean change: 0.065

### A = 42 px




max field shift: 39.0 px | mean change: 0.086

## HOW IT WORKS

The field moves visible content up and down. It keeps page edges fixed.

Use this mode to test changes in line height and baseline position.

**LIMIT:** At high intensity, text lines can cross.

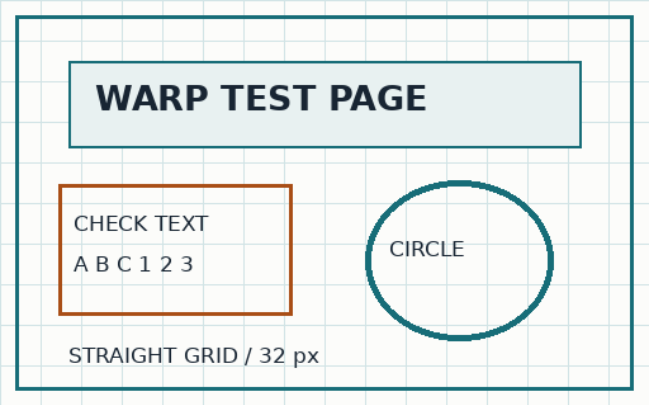
```
WarpTransform(mode="vertical_wave", intensity=18)(image)
```

EFFECT SCALE / GENTLE_PINCH

# Gentle pinch

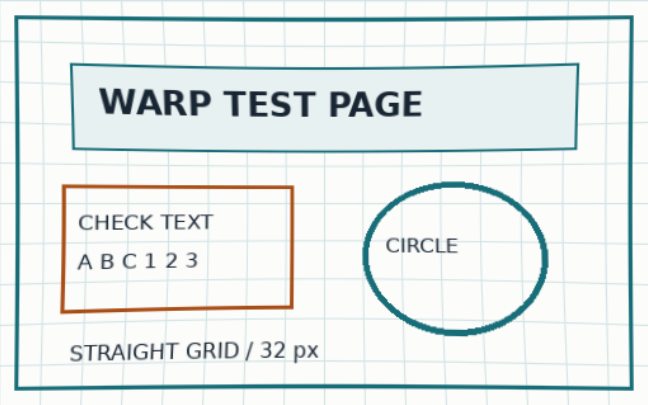
Intensity changes one fixed field.

### A = 0 px



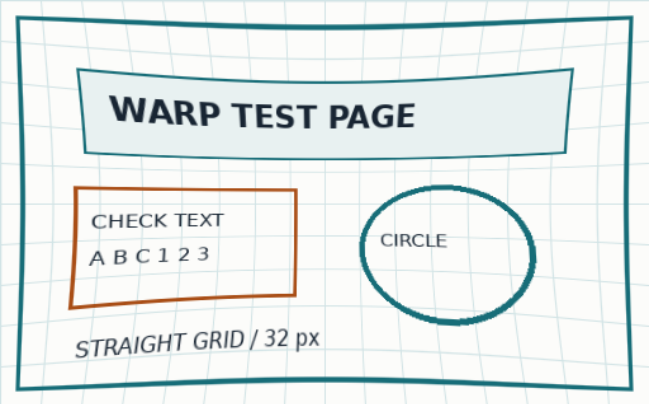

max field shift: 0.0 px | mean change: 0.000

### A = 30 px



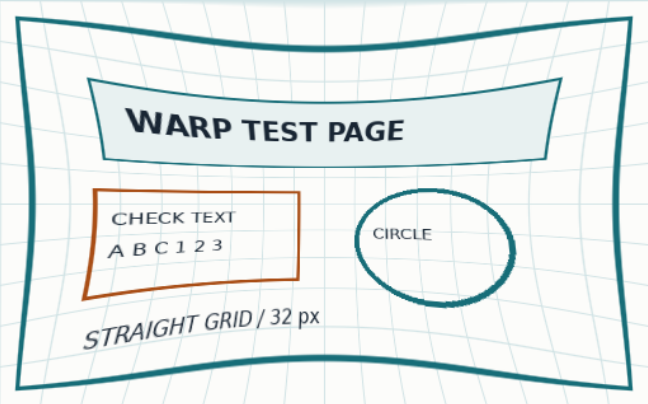

max field shift: 5.6 px | mean change: 0.078

### A = 90 px




max field shift: 16.7 px | mean change: 0.113

### A = 180 px




max field shift: 33.4 px | mean change: 0.122

## HOW IT WORKS

The field pulls visible content toward the page center. It is zero at the edges.

Use this mode to test a broad, centered geometric change.

**LIMIT:** At high intensity, local detail can be difficult to read.

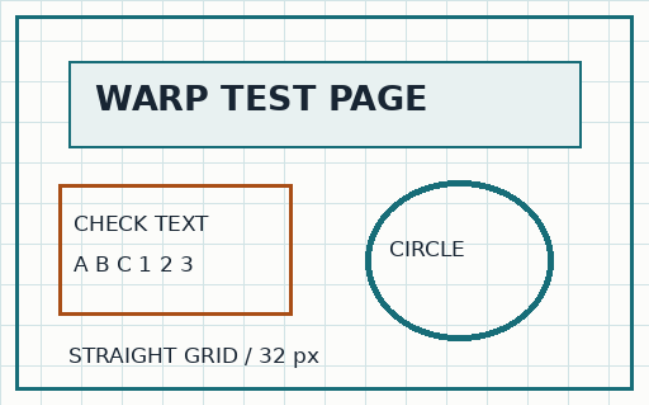
```
WarpTransform(mode="gentle_pinch", intensity=90)(image)
```

EFFECT SCALE / SINE_LENS

# Sine lens

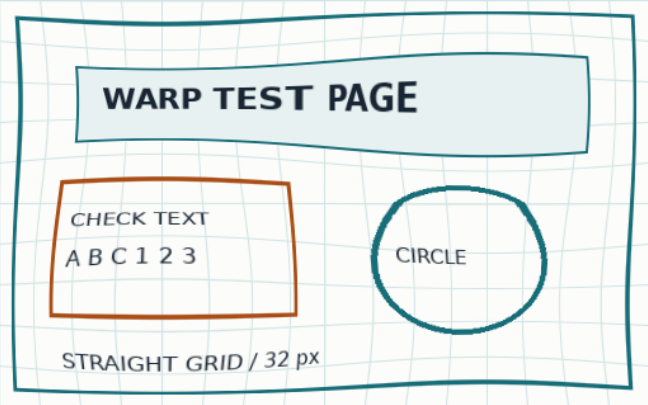
Intensity changes one fixed field.

### A = 0 px



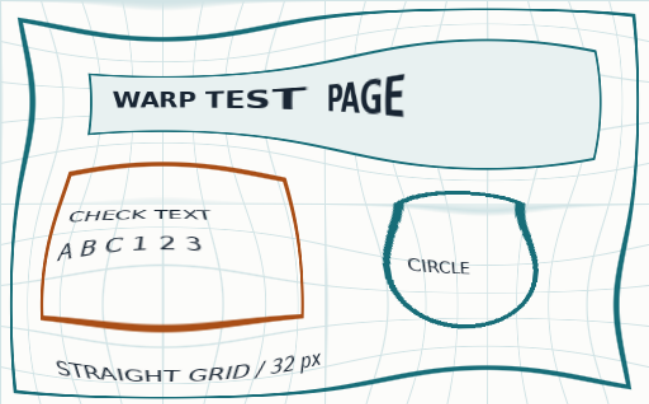

max field shift: 0.0 px | mean change: 0.000

### A = 40 px



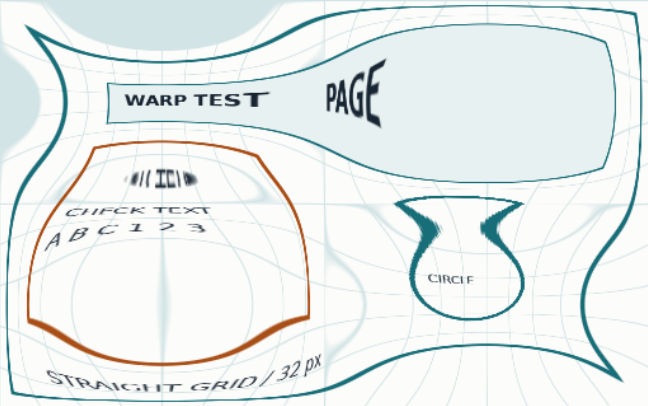

max field shift: 7.4 px | mean change: 0.095

### A = 100 px




max field shift: 18.5 px | mean change: 0.110

### A = 220 px




max field shift: 40.8 px | mean change: 0.107

## HOW IT WORKS

A repeated geometric field forms two columns and two rows of lens cells.

Use this mode to test repeated local changes across a page.

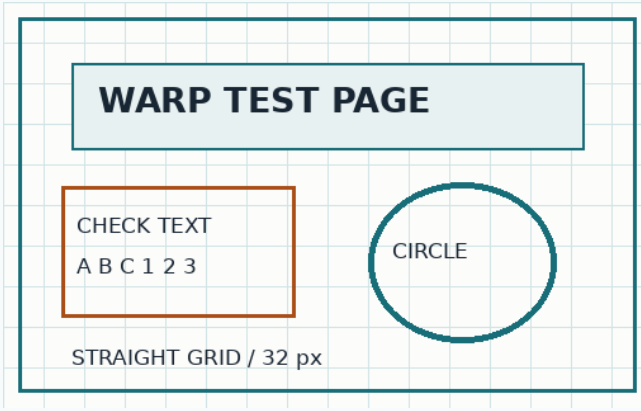
**LIMIT:** At high intensity, cells can fold or duplicate text.

```
WarpTransform(mode="sine_lens", intensity=100)(image)
```

# Cosine droplets

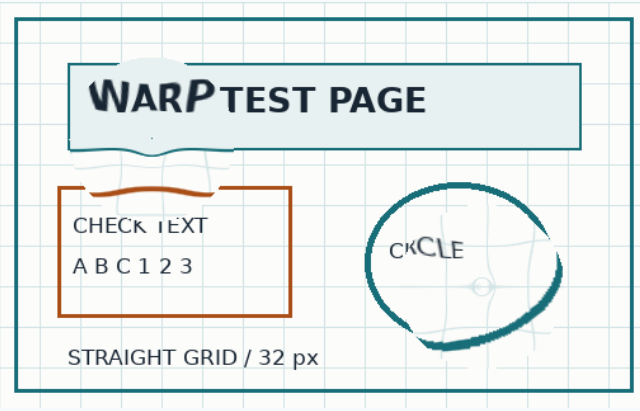
Intensity changes one fixed field.

### A = 0 px



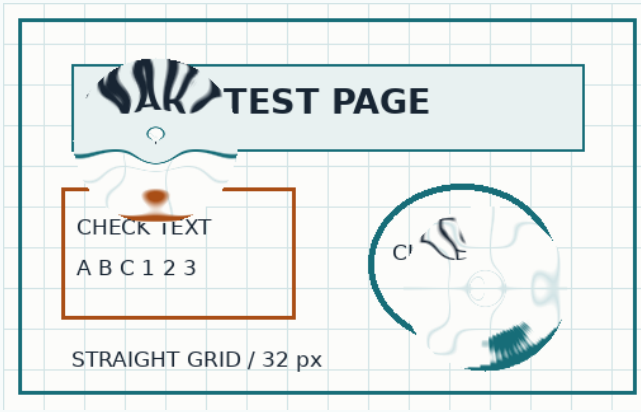

max field shift: 0.0 px | mean change: 0.000

### A = 4 px



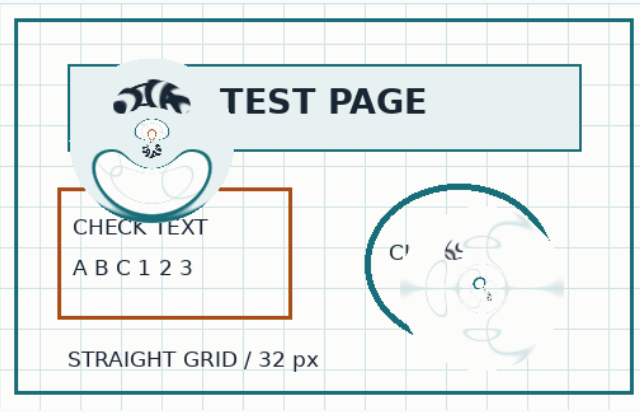

max field shift: 8.0 px | mean change: 0.013

### A = 12 px




max field shift: 24.0 px | mean change: 0.020

### A = 28 px




max field shift: 56.0 px | mean change: 0.016

## HOW IT WORKS

Two radial fields move content near fixed centers. Each field has a 65 px radius.

Use droplets to test local changes without a page-wide field.

**LIMIT:** At 28 px, local text is not legible. Border samples can repeat.

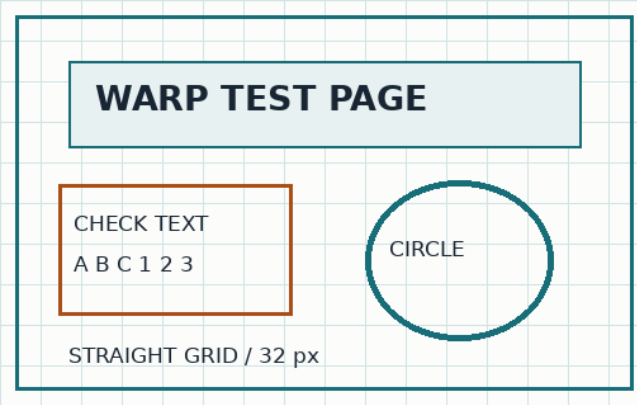
```
distort_image(image, DistortionConfig(radius=65, intensity=12), centres)
```

EFFECT SCALE / PARABOLIC

# Parabolic droplets

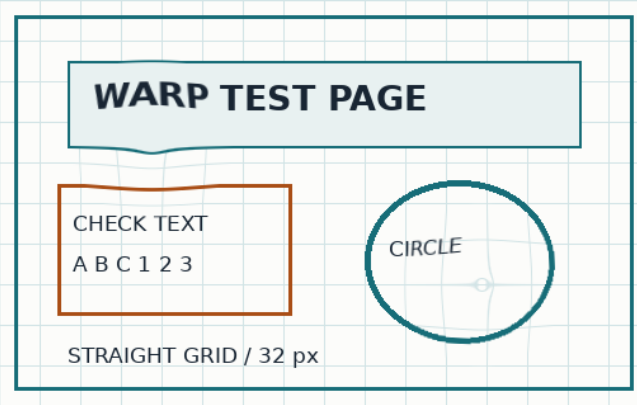
Intensity changes one fixed field.

### A = 0 px



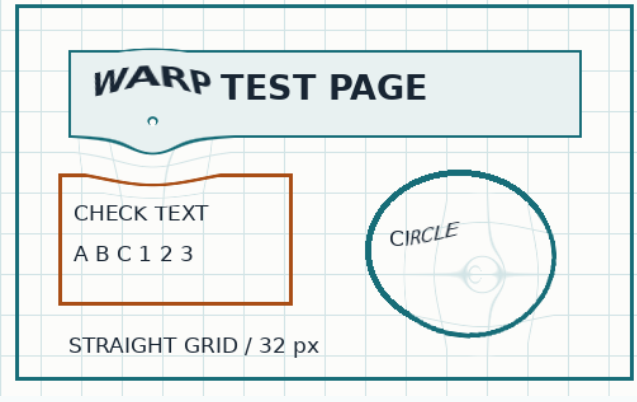

max field shift: 0.0 px | mean change: 0.000

### A = 5 px



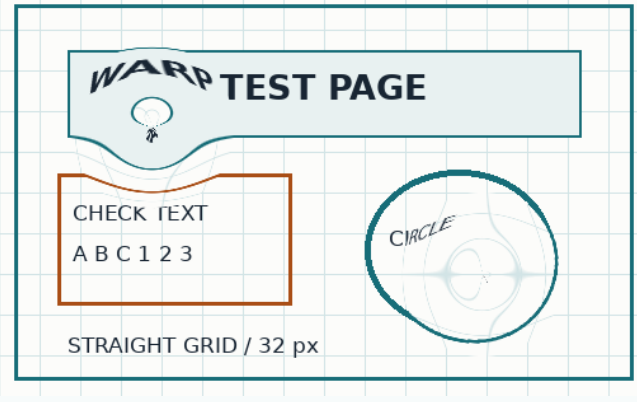

max field shift: 5.0 px | mean change: 0.011

### A = 15 px




max field shift: 15.0 px | mean change: 0.017

### A = 35 px




max field shift: 35.0 px | mean change: 0.019

## HOW IT WORKS

Two nonoverlapping radial fields decrease to zero at a 65 px radius.

Use this profile when droplet interiors must not overlap.

**LIMIT:** The API rejects overlapping parabolic centers.

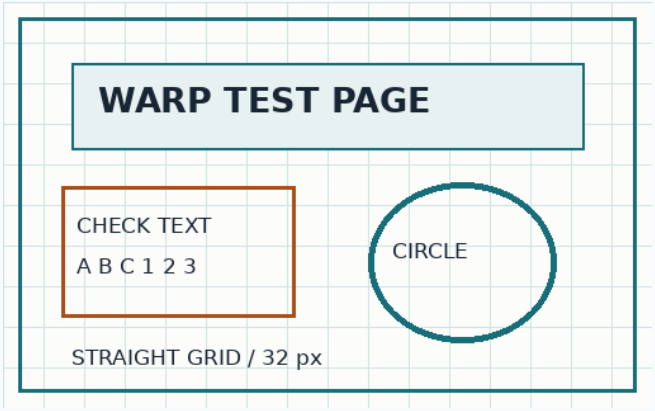
```
distort_image(image, DistortionConfig(radius=65, intensity=15, profile="parabolic"), centres)
```

EFFECT SCALE / REFRACTION

# Uniform-slab refraction

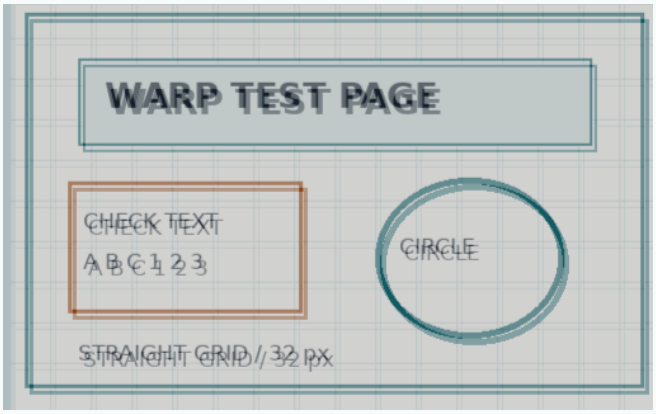
Thickness changes the physical translation.

### t = 0 mm



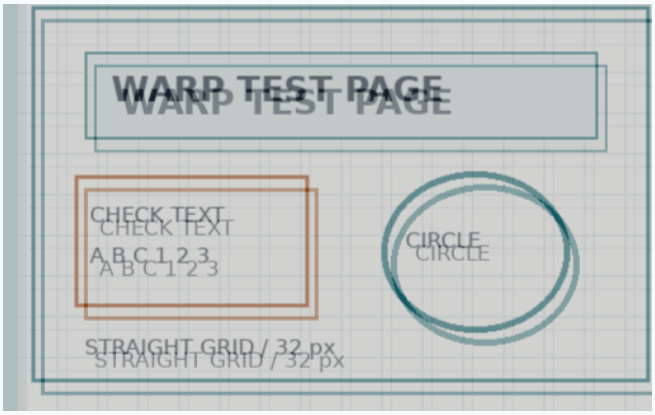

max field shift: 0.0 px | mean change: 0.000

### t = 5 mm



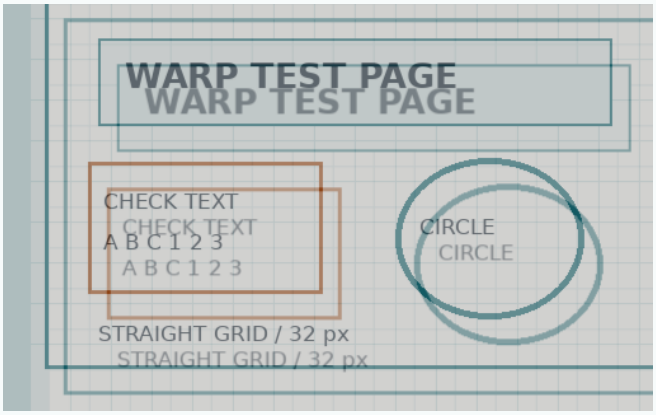

max field shift: 8.9 px | mean change: 0.219

### t = 10 mm




max field shift: 17.9 px | mean change: 0.222

### t = 20 mm




max field shift: 35.8 px | mean change: 0.224

## HOW IT WORKS

A physical thickness changes two mode translations. This is not a geometric intensity.

Use this mode to compare polarization views of a uniform slab.

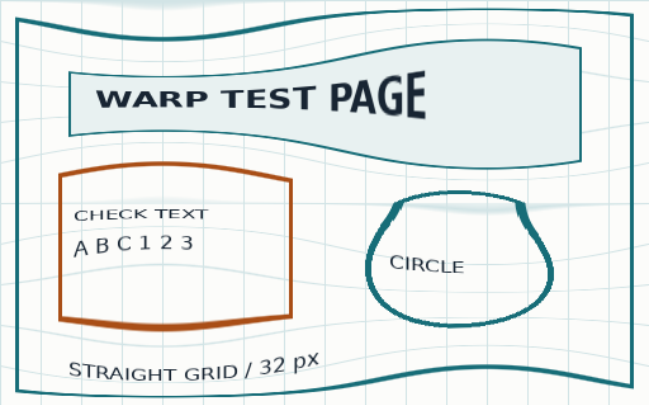
**LIMIT:** Mean change includes transmission loss. Mode brightness is approximate.

```
refract_image(image, thickness_mm=10, page_width_mm=210, page_height_mm=131.25)
```

# Change the lens shape

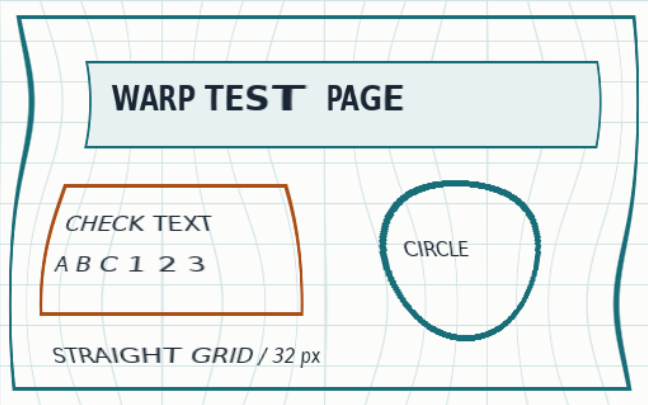
All four images use A = 100 px. Only one lens option changes per image.

### X gain = 0



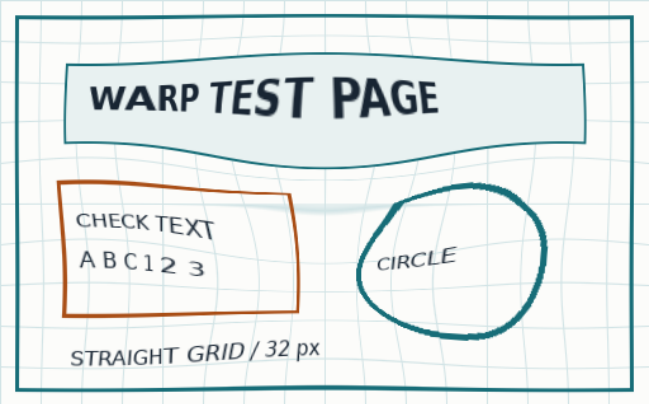

Remove X displacement.

### Y gain = 0



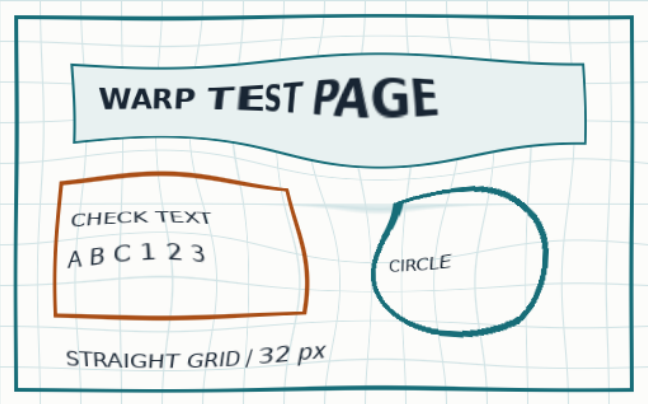

Remove Y displacement.

### X phase = 0.25




Shift X by a quarter-cycle.

### X frequency = 2.5




Use a fractional X frequency.

## COUNT AND GAIN

A sets columns. B sets rows. Each axis gain changes only its field component.

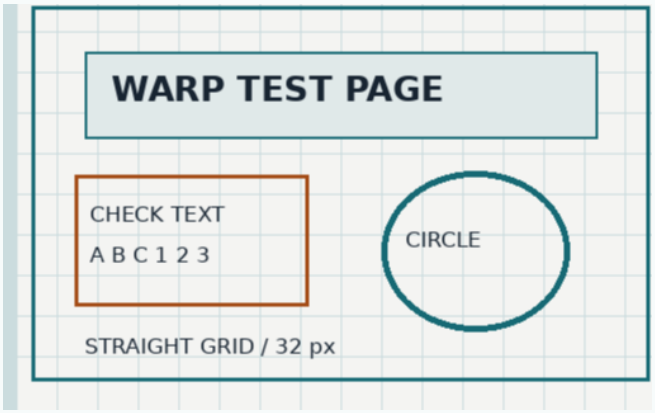
A fractional frequency or a nonzero phase adds an edge taper.

```
WarpTransform(mode="sine_lens", intensity=100,
              lens_options={"frequency_x": 2.5})(image)
```

# Separate optical views

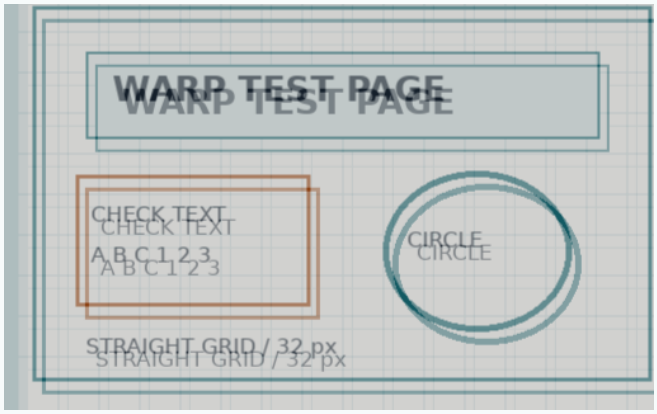
The thickness is 10 mm in all three images. Only polarization changes.

### Polarization = 0 degrees



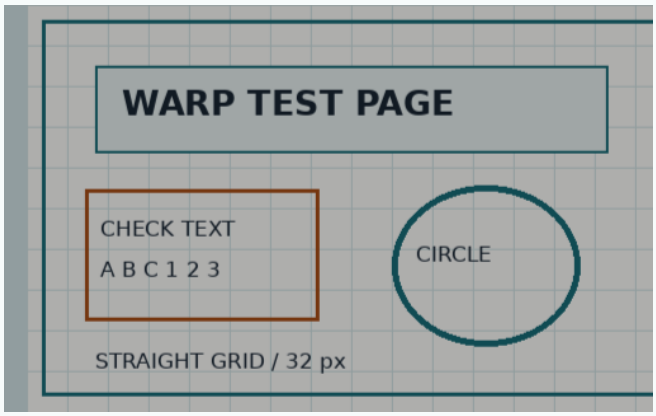

Mode weight changes; the slab stays fixed.

### Polarization = 45 degrees




Mode weight changes; the slab stays fixed.

### Polarization = 90 degrees




Mode weight changes; the slab stays fixed.

## PHYSICAL LIMITS

The slab has constant indices. It gives mode translations, not a local optical warp.

The oblique mode weights approximate transmission. The image is not an exact brightness prediction.

# Use the package

Use one CHW tensor in Python, or use one image file in the CLI.

## PYTHON: APPLY A PAGE WARP

```
import torch
from lorem_warped_examples import WarpTransform
image = torch.rand(3, 320, 512)
out = WarpTransform(mode="horizontal_wave", intensity=18)(image)
```

## PYTHON: APPLY DROPLETS

```
from lorem_warped_examples import DistortionConfig, distort_image
centres = torch.tensor([[108., 120.], [224., 380.]])
config = DistortionConfig(radius=65, intensity=12)
out = distort_image(image, config, centres)
```

## CLI: USE AN IMAGE FILE

```
lorem-warp --input source.png --output result.png \
  --page-warp sine_lens --intensity 100
```

## CLI: PRINT PHYSICAL MATRICES

```
lorem-warp --page-warp refractive_index \
  --refraction-matrices-mm
```

The CLI uses 12 px by default for geometric modes. The Python page transform uses 3 px by default.

Set intensity explicitly when you compare the CLI and Python results.

# Method and limits

This manual uses measured results, not drawings of possible output.

### INPUT

All examples use the same 512 × 320 source image and grid.

### GEOMETRIC SCALE

Each row uses one fixed field and the shown intensity in pixels.

### OPTICAL SCALE

The optical raster is 210 × 131.25 mm. X and Y use 2.44 pixels per mm.

### PIXEL CHANGE

Mean change is the mean absolute difference from the source, in [0, 1].

### OFFSET

Maximum shift is the largest field length. It is not a text-legibility score.

### READABILITY

High intensity can fold coordinates, duplicate letters, or clamp border areas.

A stronger field can show a smaller mean change after folds or border sampling.

### WRITING STANDARD

This manual follows selected ASD-STE100 Issue 9 rules for short and direct text.[1]

It has no complete dictionary review. Do not treat it as certified STE documentation.

### SOURCE

[1] ASD-STE100 Simplified Technical English, Issue 9 (2025).

www.asd-ste100.org/assets/files/ASD-STE100_ISSUE9.pdf

### PACKAGE FILES

docs/pdf_assets/manifest.json records every value and image path.

docs/build_manual.py makes this PDF from the installed package.